

REED PUTNAM PRACTICE · FALL 2026

Induction, Pigeonhole, & the Extremal Principle

Week 1 · three ways to prove something exists

Thu Sep 24 / Fri Sep 25

The toolbox (1)

Pigeonhole

N objects in n boxes put at least $\lceil N/n \rceil$ in one box. Averaging form: if $y_1 + \dots + y_n = s$, some $y_i \geq s/n$ and some $y_j \leq s/n$.

- > The art is choosing the boxes: residues mod n , odd parts, partial sums, fractional parts in short intervals, small cells of a region.
- > Ask: what do I want two objects to share?

The toolbox (2)

- > **Extremal principle**: a finite set has a largest and a smallest element. Pick the extreme object (longest, closest, least counterexample) and use that **nothing beats it**. “Sort the numbers” is this in disguise.
- > **Induction**: ordinary, strong, or on a **well-chosen parameter** ($p + q$ for p/q , the largest element, the number of odd terms). It is how “repeat this step” becomes a proof.
- > **Strengthen the hypothesis** when the step won't close:
 - $\sum_{k \leq n} 1/k^2 < 2$ resists; $\sum_{k \leq n} 1/k^2 \leq 2 - 1/n$ goes through in one line.

Example 1

Pigeonhole with partial sums

Among any n integers $a_1, \dots, a_n$, some nonempty block of consecutive terms has sum divisible by n .

→ work it on the board (3 min)

Example 2

Extremal principle in a tournament

In a round-robin tournament (every pair plays once, no ties) there is a player K such that every other player either lost to K or lost to someone who lost to K .

→ work it on the board (4 min)

Example 3

Strong induction (Putnam 2005 A1)

Every positive integer is a sum of one or more numbers of the form $2^r 3^s$ ($r, s \geq 0$) in which no summand divides another.

→ work it on the board (5 min)

How to recognize it

- > “Among any N , some two...” $\Rightarrow$ **pigeonhole**: a count one more than a modulus.
- > Must exist in a finite configuration, no formula $\Rightarrow$ **extremal**: largest, closest, longest.
- > Indexed by n , smaller case inside $\Rightarrow$ **induction**; not $n - 1$? strong, or a new parameter.

Pitfalls

- > Pigeonhole gives *some* two in a box, not the two you wanted.
- > “Smallest counterexample” needs a well-ordered set; say why the extreme object exists.
- > State $P(n)$ exactly; check the base case; don’t silently assume $n \geq 2$.

Writing a solution that scores

- > Say precisely what you will prove, including any reformulation.
- > Full sentences, **every quantifier visible**; equations without words are not a proof.
- > Justify every step a skeptic could question, **naming the tool** as you use it.
- > Scoring: 10 for complete, 8–9 for a small gap, **almost always 0–2 otherwise**.
- > Cross out what you abandon; **never claim more than you have shown**.

Now you

Problems 1–4.

1–2 warm-ups · 3–4 easy Putnam · 5–6 take-home

35 minutes, groups of 2–3

Solutions posted after Friday.